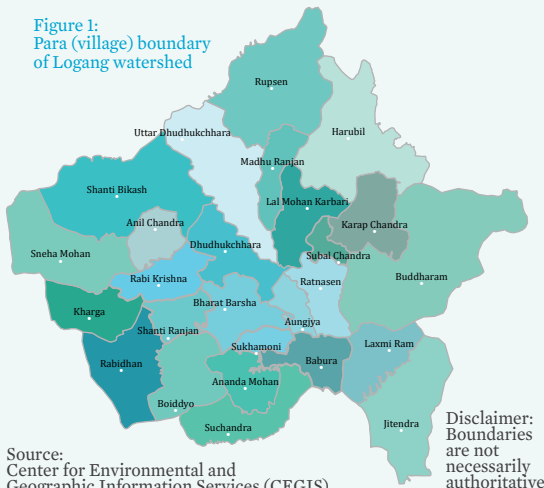


Logang Watershed

K-98



Figure 1: Para (village) boundary of Logang watershed



Source: Center for Environmental and Geographic Information Services (CEGIS)

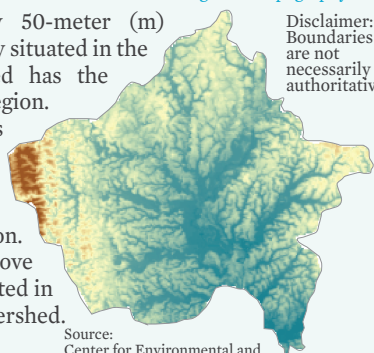
Disclaimer: Boundaries are not necessarily authoritative

The Logang sub-watershed (K-98), located in the 02 no Chengi and 01 no Logang Union under the Panchhari Upazila of Khagrachhari District, encompasses 26 paras (figure 1) and covers approximately 2,982 hectares of land. It is home to over 7,000 people, with communities consisting of the Chakma, Marma, and Tripura ethnic groups. The overall literacy rate is approximately 89%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 15,178 taka (\$126), which is relatively low compared to the national average. About 84% of the houses are katcha and almost 64% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 16% area lies below 50-meter (m) elevation (figure 2). The valley situated in the center of the sub-watershed has the lowest elevation in the region. Most of the study area lies between 75-90m (36%) and 90-105m (38%) elevation. About 7% of the area lies between 105-120m elevation. Only 3% of the area lies above 120m elevation which is situated in the western region of the watershed.

Figure 2: Topography



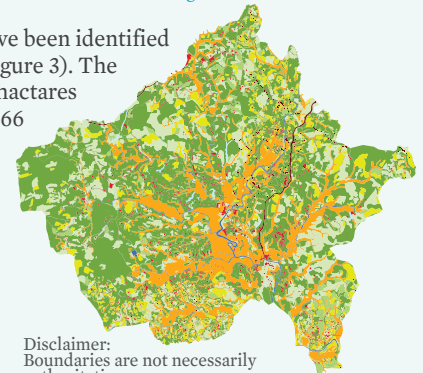
Source: Center for Environmental and Geographic Information Services (CEGIS)

Disclaimer: Boundaries are not necessarily authoritative

Landuse

A total 13 types of land use have been identified within the K-98 watershed (figure 3). The main land use is forest (1,367 hectares (ha)), followed by cropland (566 ha), herb/shrub (488 ha), and shifting cultivation (280 ha) mainly in valleys and slopes.

Figure 3: Land use

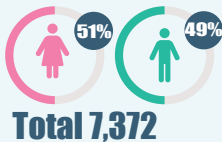


Disclaimer: Boundaries are not necessarily authoritative

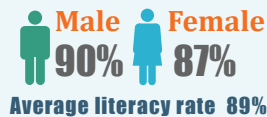
Source: Center for Environmental and Geographic Information Services (CEGIS)

Demography

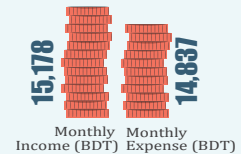
Population



Literacy



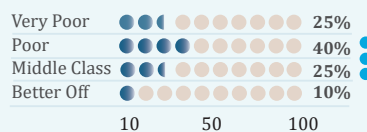
Economy



WASH



Poverty



Production

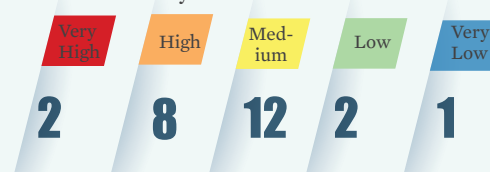


Climatic Hazards



Climate Change-Induced Hazards

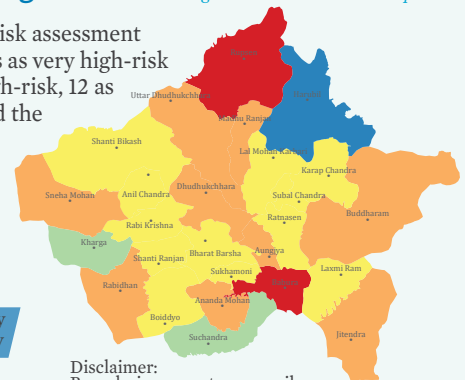
Nine climate hazards threaten Logang watershed's livelihoods and biodiversity. Drought and heat waves are intensifying across most paras. Erratic rainfall, lightning, river floods, and flash floods are increasing in frequency and severity, ranking highest in impact. Hail storms, cold waves, and landslides are occurring at low to moderate intensity.



Climate Change Risk

Climate change risk assessment identified 2 paras as very high-risk hotspots, 8 as high-risk, 12 as medium-risk, and the remaining paras as low to very low-risk (figure 4).

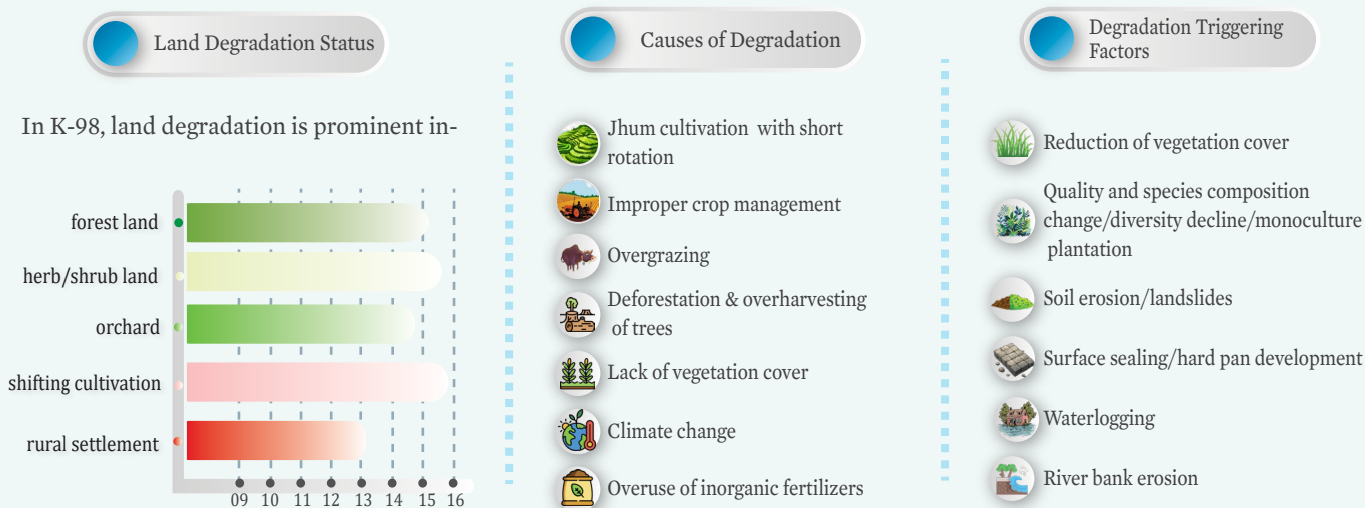
Figure 4: Multi-hazard risk map



Disclaimer: Boundaries are not necessarily authoritative

Source: Center for Environmental and Geographic Information Services (CEGIS)

Watershed Degradation

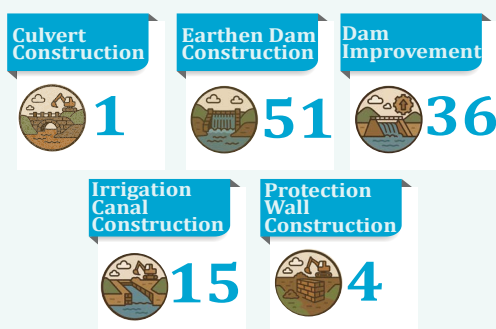


Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.



Watershed Management Infrastructures



Watershed conservation interventions in Logang include dam construction (46) and dam improvement (36) being the most implemented, followed by irrigation canal construction (15), aimed at enhancing water storage and distribution. In addition to these, other watershed conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Logang region.